

The recursive block decomposition of the CRT split

A proven torsion-localization, a validated per-block rank law,
and a sofa-computable block count for Fermat cells

Rafael Amichis Luengo

Madrid • github.com/tretoef-estrella

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Abstract

This note is the companion to *The prime-power reduction frontier* (`THE_FRONTIER_6_6.md`). The Frontier explained why a fast cohomology reduction lies on composite degree—the cyclotomic polynomial factors over $\mathbb{F}_p$ —and its constructive dual, the *CRT block split*, turned that same factorization into a memory lever. Here we report the exact structure found *inside* that split. Three results are given in descending order of how firmly they stand: (i) **proven**—torsion localizes to a single CRT block, by centrality of the idempotents; (ii) **strongly validated, out of sample, with one explicit open residual**—each block’s rank is a product of two Degtyarev–Shimada recursion ladders, each a reduced-dimension Fermat rank; (iii) **verified byte-exact**—the number of blocks of a cell is computable in closed form from elementary cyclotomic number theory. The ordering is deliberate: nothing is dressed above its evidence, and the single open problem—that the CRT idempotent cut equals the Degtyarev–Shimada §17 coordinate cut—is stated precisely. The torsion counterexample (the *swan*) has not been found; every measured block is primitive. What the decomposition adds is an explicit recursive anatomy of these cells’ integral cohomology together with a proof that any torsion is confined to one computable block—the per-block elementary-divisor structure that Aljovin–Movasati–Villaflor explicitly left uncomputed.

How to read this. Section 1 fixes notation. Section 2 is a self-contained proof—torsion localizes to one block—and may be read alone. Section 3 is the main result: a per-block rank law, validated out of sample, with its single open residual stated explicitly in §3.2. Sections 4–5 give the block count and its conjectural extension. Section 6 is the honest scope and the graveyard of refuted approaches. The punchline, for a reader in a hurry: localization is *proven*; the rank law is *validated, not yet proven*; the swan has not been found.

1. Notation and setting

Fix a Fermat cell (n, m) with $n = 2d$ even, and a prime $p \mid m$. Over $\mathbb{F}_p$ the polynomial $t^m - 1$ (equivalently the relevant φ) factors; grouping the $(t - 1)$ -primary part as A and the remainder as B , the per-variable quotient ring splits by the Chinese Remainder Theorem. Write $\dim_A$ and $\dim_B$ for the $\mathbb{F}_p$ -dimensions of the two factors, with $\dim_A + \dim_B = m - 1$, and $NV = n + 1$ for the number of variables. The cohomology module splits into blocks indexed by which of the NV variables sit in A and which in B .

For a two-component cell $(A \times B)$ —the case in which every measured cell of this campaign lives) there are 2^{NV} blocks, indexed by a binary mask of length NV ; the number of B -variables in a block is its *popcount* k . The general component count is given in Section 4.

Throughout, $DS_n(m)$ denotes the Degtyarev–Shimada closed-form rank of $L(X)$ for the Fermat cell of dimension n and degree m (Remark 4.4 of [1]). This is the *closing rank* $= \text{DIM} - \dim_{\mathbb{C}}$, equal to the quantity the prime-field reduction engine sums to; it is not $\dim_{\mathbb{C}}$ itself. All ranks below are in this sense.

2. Result one (proven): torsion localizes to a single block

Theorem 1 (Localization). *For every cell and every prime $p \mid m$, the integral cohomology module M decomposes as a direct sum over the CRT blocks. Consequently any torsion subgroup of M is confined to a single block—the unique block whose rank drops modulo p .*

Proof. The CRT idempotents e_i effecting the split are polynomials in the shift variable t : they are the standard idempotents $\prod_{j \neq i} (t - r_j) / (r_i - r_j)$ lifted from the factorization of the minimal polynomial. A polynomial in t commutes with multiplication by t ; since the module structure is generated by the shift, each e_i is *central*. The idempotents are orthogonal ($e_i e_j = 0$ for $i \neq j$) and complete ($\sum_i e_i = 1$)—the defining CRT relations. A central, orthogonal, complete system of idempotents yields a direct-sum decomposition $M = \bigoplus_i e_i M$ into submodules. A direct sum has a block-diagonal presentation matrix, and the Smith normal form of a block-diagonal matrix is the concatenation of the Smith normal forms of its blocks. Hence the elementary divisors—equivalently, the torsion—partition across the blocks, and any torsion is localized to the block(s) carrying it. $\square$

Significance. This is the rigorous half of a torsion detector. For *every* cell, with no computation and no unproven law, it says *where* a counterexample must be: in the one block that drops rank modulo p . It requires no dictionary between our blocks and any external construction, and no transfer of an external corollary; it follows directly from the idempotents being polynomials in the shift, which they are by construction.

What it does not give. Localization says *where*, not *what*. It does not by itself give the rank a primitive block should carry (that is Result Two), nor does it constitute a tested detector (the value law it would be paired with is validated, not proven; see Section 6).

Verifier. `verifiers/CENTRALITY_DETECTOR_VERIFIER.py` constructs the idempotents for several cells (characteristic 2, 3, 5; $m = 6, 12, 15$) and checks byte-exact that they are central, orthogonal, idempotent, and complete—the hypotheses of Theorem 1.

3. Result two (strongly validated; one open residual): the per-block rank is a product of two recursion ladders

Proposition 1 (Per-block product law). *Define the recursion ladder*

$$\text{rec}(d, \ell) = \begin{cases} 1 & \ell = 0, \\ d & \ell = 1, \\ \text{DS}_{2(\ell-1)}(d+1) & \ell \geq 2. \end{cases}$$

For a block with k of the NV variables in B (so $\text{NV} - k$ in A), set $\ell_A = \lceil (\text{NV} - k)/2 \rceil$ and $\ell_B = \lceil k/2 \rceil$. Then

$$\text{rank}(\text{block}) = \text{rec}(\dim_A, \ell_A) \cdot \text{rec}(\dim_B, \ell_B).$$

The block rank is a product of two ladders—one running the recursion on the A factor with the A -variables, the other on the B factor with the B -variables. Each ladder value $\text{DS}_{2(\ell-1)}(d+1)$ is the Degtyarev–Shimada rank of a Fermat cell of *reduced* dimension $2(\ell - 1)$ and grade $d + 1$: each block is, numerically, a smaller Fermat cell.

3.1. The evidence, in order of strength

Per-block byte-exact match. The law reproduces every individual block rank—not merely totals—on all four fully dumped cells: (4, 6) char 2 ($\dim_A=1, \dim_B=4$), (4, 10) char 2 ($\dim_A=1, \dim_B=8$), (4, 6) char 3 ($\dim_A=2, \dim_B=3$), and (4, 12) char 3 ($\dim_A=2, \dim_B=9$). For instance the (4, 6) char 3 block values 20, 18, 18, 38, 38, 141 (by popcount class) are each reproduced.

Blind out-of-sample prediction. The ladder multipliers were extracted from cells with $\dim_A \in \{1, 2\}$ only. The law was then applied, unseen, to (4, 15) char 5, which has $\dim_A = 4$ —a value never used in fitting. It predicted the block ranks by popcount as 400, 360, 360, 1080, 1080, 10900; the measured distinct values 360, 1080, 10900 match. A curve fit cannot extrapolate to a new $\dim_A$; a product of fixed recursion ladders does, because the A -side multiplier is not a free per- $\dim_A$ parameter—it is the same $\text{rec}(\cdot)$ evaluated at $\dim_A$. This is the decisive evidence that the structure is real and not a fit.

A measured high-level monster. The cell (4, 14) char 2 ($\dim_A=1, \dim_B=12$) closed PRIMITIVE on the 8 GB laptop at peak 2.218 GB, and its largest block (popcount 5, hence $\ell_B = 3$) measured $19,920 = \text{DS}_4(13)$ byte-exact—the first directly measured high- ℓ block at this $\dim_B$, confirming the ladder value with measured data rather than deduction.

3.2. The single open residual

What is *proven* is the *form* (a product of two ladders), confirmed past fitting by the blind prediction. What is *measured but not yet proven in general* is the ladder *value*: that $\text{rec}(d, \ell) = \text{DS}_{2(\ell-1)}(d+1)$. The numbers match the Degtyarev–Shimada §17 construction byte-exact (ratio 1.000, every block, multiple cells), but equal numbers are not yet a proof that the objects coincide. Section 17 of [1] builds an explicit sub-Fermat by zeroing coordinates,

$$X(2s) := W_s \cap \{z_{2s+2} = \cdots = z_{n+1} = 0\},$$

a Fermat variety of dimension $2s$, and counts $(2s+1)!! \cdot m^{d+1}$ linear subspaces in it. Our block of recursive dimension $2(\ell-1) = 2s$ carries the same rank. The open problem—pen and paper, needing no further computation—is:

Prove that the CRT idempotent projection (the $(t-1)$ -vs- $g(t)$ ring split) realizes the Degtyarev–Shimada coordinate-zeroing cut $W_s \cap \{z = 0\}$; that is, that our block and $X(2s)$ are the same reduced-dimension sub-Fermat, not merely two objects of equal rank.

Proving this converts Proposition 1 from a validated law into a theorem and, paired with Theorem 1, yields a *certified* detector: a measured block whose rank departs from $\text{DS}_{2(\ell-1)}(d+1)$ in a recursive-dimension- ≥ 4 block would be a provable torsion counterexample.

Verifiers. `GENERAL_PRODUCT_LAW_VERIFIER.py` checks the product law per block on the four dumped cells and reproduces the blind (4, 15) char-5 prediction; `DS_CONNECTION_VERIFIER.py` checks the reduced-dimension/grade identification against §17; `TRIPLE_TIE_VERIFIER.py` checks the double-factorial fingerprint below.

3.3. The triple tie: one fingerprint, three faces

The double factorial $(2s+1)!! = 1, 3, 15, 105$ appears in three places that the recursion shows to be one:

1. the leading coefficient of DS_{2s} (Remark 4.4): $\text{DS}_2 \sim 3m^2$, $\text{DS}_4 \sim 15m^3$, $\text{DS}_6 \sim 105m^4$;

2. the subspace-count prefactor in §17: W_s contains $(2s + 1)!! \cdot m^{d+1}$ linear subspaces;
3. the multiplicity backbone of the recursive-dimension blocks (the binomial palindrome below).

That the same double factorial governs all three is the fingerprint that the CRT block, the DS_{2s} value, and the count of s -subspaces in $X(2s)$ are the same object—strong corroboration of the identification the residual asks to prove, though corroboration is not the proof.

3.4. Separability and the binomial palindrome

Two structural facts fall out of the form, both verified.

Separability. The block rank depends on $(\dim_B, c)$ only, not on the mother dimension n . Measured: $(6, 10)$ char 2 ($n=6, \dim_B=8$) and $(4, 10)$ char 2 ($n=4, \dim_B=8$) produce identical low-block values 1, 8, 8, 168, 168. The mother dimension sets only how many recursion rungs exist (up to recursive dimension n); the rung values are shared—exactly what the §17 tensor predicts, the low-dimension factor being independent of the total dimension.

Palindrome. Grouping blocks by recursion level, the multiplicities are palindromic: $n=4$ gives 1, 15, 15, 1; $n=6$ gives 1, 28, 70, 28, 1; $n=8$ gives 1, 45, 210, 210, 45, 1. Each sums to 2^{NV} and is symmetric under $c \leftrightarrow (\text{top} - c)$. The pairing observed from the start—adjacent levels carrying equal weight—is this binomial palindrome.

4. Result three (verified byte-exact): the block count is computable from the sofa

The $A \times B$ (two-component) picture is a special case. In general $t^m - 1$ factors over $\mathbb{F}_p$ into more than two coprime primary components, and the per-variable ring splits into that many factors.

Proposition 2 (Block-count law). *Let m' be the prime-to- p part of m . The number of coprime primary components per variable is*

$$1 + \sum_{\substack{d|m' \\ d>1}} \frac{\varphi(d)}{\text{ord}_d(p)},$$

where φ is Euler's totient and $\text{ord}_d(p)$ is the multiplicative order of p modulo d . The total number of blocks of a cell is this count raised to the power NV .

The “1” is the $(t - 1)$ -primary part; each divisor $d > 1$ of m' contributes $\varphi(d)/\text{ord}_d(p)$ irreducible factors of the d -th cyclotomic polynomial, each a primary component.

Evidence. Verified byte-exact against direct factorization of $t^m - 1$ over $\mathbb{F}_p$ for all $p \in \{2, 3, 5, 7, 11\}$ and $m \in [3, 24]$ with $p \mid m - 28$ cases, zero mismatches. The count is frequently more than two: $(4, 14)$ char 2 $\rightarrow 3$; $(4, 18)$ char 2 $\rightarrow 3$; $(4, 12)$ char 3 $\rightarrow 3$; $(4, 24)$ char 3 $\rightarrow 5$; $(4, 20)$ char 5 $\rightarrow 4$; $(4, 21)$ char 7 $\rightarrow 3$.

Consequence for the hunt. With Theorem 1 (torsion in one block), this gives from the armchair the size of the swan's hiding space per cell—how many blocks the counterexample could occupy—before any computation. $(4, 14)$ char 2 has $3^5 = 243$ rooms; $(4, 6)$ char 2 has $2^5 = 32$. Richer prime structure means a finer lattice with more, smaller rooms.

A correction this carries. The “ 2^{NV} masks” framing used while the decomposition was first mapped is the two-component special case. Every cell *measured* in this campaign is two-component, so the per-block tables and the validated law of Section 3 stand exactly as stated; but the general detector must use the count above. The > 2 -component law is the conjecture of Section 5.

Verifier. `BLOCK_COUNT_LAW_VERIFIER.py`.

5. The r-ladder: a flagged conjecture for more than two components

Conjecture 1 (r-ladder). *For a cell whose per-variable ring has r coprime primary components $C_1, \dots, C_r$, a block (indexed by the distribution $n_1 + \dots + n_r = \text{NV}$ of the variables among the components) has rank $\prod_{i=1}^r \text{rec}(\dim_i, \lceil n_i/2 \rceil)$ —a product of r ladders, one per component.*

This is a conjecture and is flagged as such. There is no measured > 2 -component cell in hand to test it, and it carries a free ambiguity only data can resolve: the ladder’s dimension parameter could be the component’s $\mathbb{F}_p$ vector-space dimension or its Fermat- $\dim_A$ -type value (these differ—for $(4, 14)$ char 2 the $(t - 1)$ -component has vector dimension 2 but Fermat type $\dim_A = 1$). The localization of Theorem 1 holds for *all* components (its proof does not assume two); what is unverified is the per-block *value* law for $r > 2$.

The datum that would test it. The smallest virgin > 2 -component cell that is RAM-reachable on 8 GB is $(4, 14)$ char 2 split into its true three components (the $(t - 1)$ part and the two cubic factors of $t^7 - 1$ over $\mathbb{F}_2$). The current dump engine groups the two cubics into a single B block (running the cell as two-component), so the existing $(4, 14)$ run does *not* test the conjecture; an engine that splits B into its two cubic blocks (a $3^5 = 243$ -block lattice) is required. This is logged as the next measured target.

6. Honest scope

- **Proven:** torsion localizes to a single CRT block, for every cell (Theorem 1).
- **Strongly validated out of sample, one open pen-and-paper residual:** the per-block product law (Proposition 1); the residual is to prove the CRT cut equals the §17 coordinate cut.
- **Verified byte-exact:** the sofa block-count law (Proposition 2).
- **Conjectural, untested:** the r-ladder generalization to > 2 -component cells (Section 5).

Detector status. Localization is the *where*; the validated law is the *what*. A certified detector needs both, and needs the residual proven—because a law fit to primitive cells predicts “primitive” by construction, so only a proof of the value law makes a measured departure mean torsion rather than a gap in the law. Until then, no nonzero scar would be trusted as a real torsion verdict without a synthetic positive control firing correctly first.

The graveyard (dead veins, recorded so they are not retried). *The partition-mask dictionary.* An early plan to certify the law by mapping the Degtyarev–Shimada coordinate-partitions J (count $(2d + 1)!!$ —15 for $n=4$, 105 for $n=6$) to our CRT masks (count 2^{NV} —32 for $n=4$, 128 for $n=6$) piece by piece was refuted with data: the counts misalign ($15 \neq 32$, $105 \neq 128$) and our linear layer has no Degtyarev–Shimada counterpart. The dictionary does not exist as a piecewise correspondence; the correct road to certification is the centrality proof of Theorem 1, which needs no dictionary.

The “total = DS” check is *vacuous as evidence*. Summing the product law over all masks reproduces $DS_n(m)$ for *any* split $\dim_A + \dim_B = m - 1$, including non-existent splits—a binomial identity true by construction. It proves nothing about per-block correctness. It is a necessary *consistency* check (it reflects the characteristic-independence of $\dim_{\mathbb{C}}$: different primes give different splits, all of which must sum to the same $\dim_{\mathbb{C}}$), but it is not evidence for the block structure; the real evidence is the per-block match and the blind prediction.

What this is not. This concerns the *integral* Hodge conjecture for *Fermat* varieties—a specific, bounded problem. It is not the Clay Millennium (rational) Hodge conjecture and makes no claim on it. The swan has not been found; every measured block is primitive. What the decomposition adds is an explicit recursive anatomy of these cells’ cohomology, a proof that any torsion is confined to one computable block, and a from-the-sofa map of where to look and what number would betray a counterexample—the per-block elementary-divisor structure that Aljovin–Movasati–Villaflor [2] explicitly left uncomputed.

7. Reproducibility

Each result has a self-contained Python verifier that recomputes the Degtyarev–Shimada values from scratch:

```
python3 verifiers/CENTRALITY_DETECTOR_VERIFIER.py # Thm 1: idempotents central; module spl
python3 verifiers/GENERAL_PRODUCT_LAW_VERIFIER.py # Prop 1: product law + blind dimA=4
python3 verifiers/DS_CONNECTION_VERIFIER.py # Prop 1: reduced-dim/grade vs DS se
python3 verifiers/TRIPLE_TIE_VERIFIER.py # the double-factorial fingerprint
python3 verifiers/BLOCK_COUNT_LAW_VERIFIER.py # Prop 2: block count vs factorizati
python3 verifiers/RIFLE_FINAL.py 4 35 # the sofa predictor, [FIRM]/[CONJ]
```

`RIFLE_FINAL.py` is the from-the-sofa predictor: it states a primitivity verdict and $\dim_{\mathbb{F}_p}$ for any cell (n, m) , $n \in \{4, 6\}$, for every prime $p \mid m$, flagging each prime [FIRM] (two-component, the validated law applies) or [CONJ] (more than two components, resting on the unverified r-ladder). The flag is the discipline: a conjectural prediction is never printed as a confirmed one.

References

- [1] A. Degtyarev, I. Shimada, *On the topology of projective subspaces in complex Fermat varieties*. J. Math. Soc. Japan **68**:3 (2016), 975–996. doi:10.2969/jmsj/06830975. arXiv:1405.4683. (The criterion; §17, the coordinate construction $X(2s)$; Remark 4.4, the closed-form rank.)
- [2] E. Aljovin, H. Movasati, R. Villaflor, *Integral Hodge conjecture for Fermat varieties*. J. Symbolic Computation **95** (2019), 177–184. arXiv:1711.02628. (Tabulates the primitive lattice but leaves its elementary divisors uncomputed.)